

Exercise 1: Sample Bias

Assume that you contact a sample of people with different score of extraversion. Assume that these scores range on a scale from 0% to 100%. Moreover, a person's probability to actually respond to the survey invitation and report their extraversion score is this score itself.

a) In any given population/sample, what is the expected actual score?

We can assume a uniform distribution of extraversion, therefore the expected mean ($\bar{x}_p$) can be calculated as $\bar{x}_p = \frac{\text{upper bound} + \text{lower bound}}{\# \text{ of bounds}}$

in our case: $\frac{100\% + 0\%}{2} = 50\%$

b) In contrast to this, which average score can you expect to find?

As the score of extraversion affects our probability to hand in, we can consider the following example. First, **for 2 respondents**:

Respondent	Score	Probability	Expected Value
1	100%	100%	100%
2	50%	50%	25%

The expected mean score for this sample is therefore $\bar{x}_s = \frac{100\% + 25\%}{100\% + 50\%} = 83.\bar{3}\%$

Now, **for n respondents**:

$$\bar{x}_s = \frac{\text{score} * \text{probability}}{\text{probability}} = \frac{\int_0^1 x * x}{\int_0^1 x} = \frac{\left[\frac{1}{3}x^3\right]_0^1}{\left[\frac{1}{2}x^2\right]_0^1} = \frac{1/3}{1/2} = \frac{2}{3} = 66.\bar{6}\%$$

Exercise 2: AI Checks

Assume that you include a control question in your survey to make sure to identify AI bots to exclude them from your data. Specifically, you use the ARC-style question below, of which you assume that $p_h = 95\%$ of human participants will actually select the correct answer (i.e., “B”) and that AI bots will not do better than chance (i.e., randomly picking the correct answer in $p_c = 25\%$ of all cases). Looking at your data, you see that $C = 70\%$ of the sample has answered the control question correctly.

a) What is your estimate for the share of AI bots in your sample?

Let b be the share of AI bots in the sample, so $(1 - b)$ is the share of humans. The observed share of correct answers C is a weighted average of the human and bot success rates:

$$C = (1 - b) \cdot p_h + b \cdot p_c$$

Solve for b :

$$C = p_h - b \cdot p_h + b \cdot p_c$$

$$C - p_h = b \cdot (p_c - p_h)$$

$$b = (p_h - C) / (p_h - p_c)$$

Insert $p_h = 0.95$, $p_c = 0.25$, $C = 0.70$:

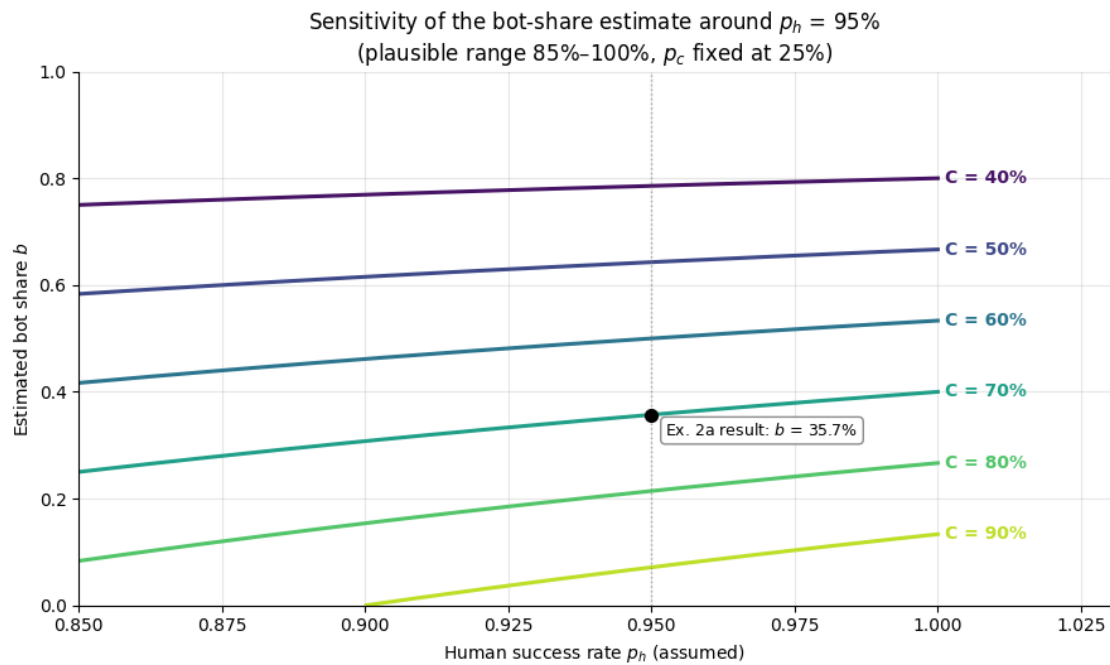
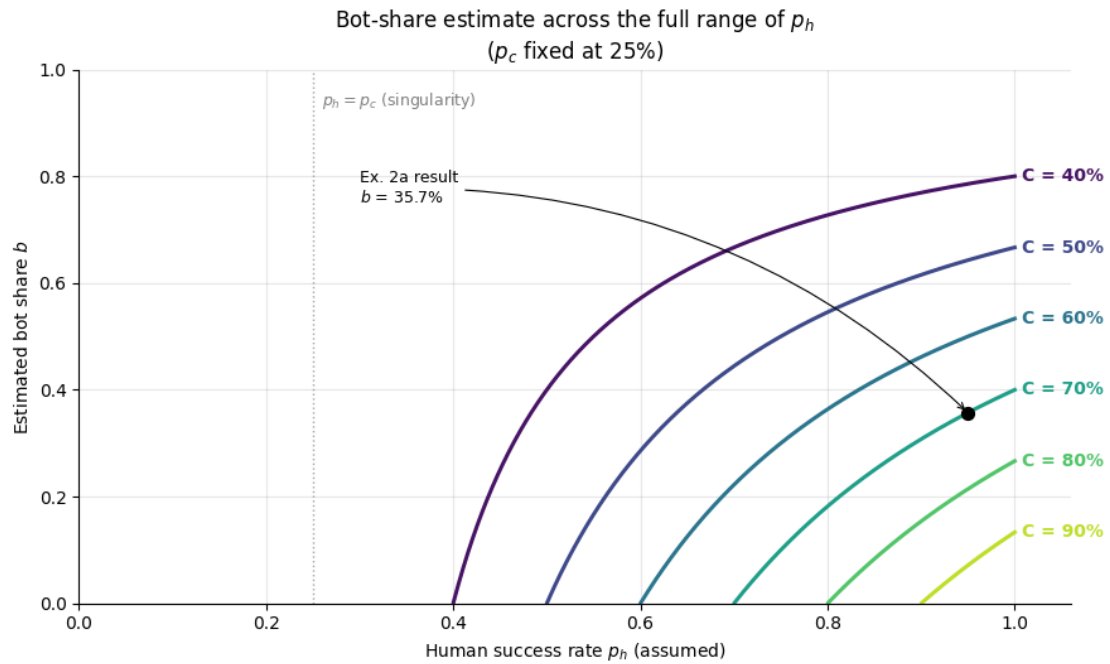
$$b = (0.95 - 0.70) / (0.95 - 0.25) = 0.25 / 0.70 \approx \mathbf{0.357}$$

So, roughly **35.7% of the sample are estimated to be bots**.

Extra:

- If $C = p_h \rightarrow b = 0$: no bots; all humans, answering correctly at the human rate
- If $C = p_c \rightarrow b = 1$: everyone performs at chance; all bots

b) Plot this result for different values of p_h (x-axis from 0 to 100%) and C (40%, 50%, ..., 90% as color).



Exercise 3: Constructs and Items

You want to conduct a survey and assess your participants' perceptions of *flow* (i.e., the mental state in which a person is fully immersed in an activity, experiencing energized focus, deep involvement, and enjoyment in the process).

- a) **Conduct a literature search on Google Scholar (or other tools) to identify a couple of established papers for this regard to identify a scale.**
- b) **Use ChatGPT (or any other AI chat tool) to create 3 to 6 items yourself.**

➔ *Try it yourself – for reference:*

AI-assisted literature search · Claude Opus 4.7 (Anthropic) · 27 April 2026 · Web search enabled · Full conversation: <https://claude.ai/share/4f4a4d09-1d0d-4d99-bda0-93662cf1571c>

Exercise 4: Networks and Causality

You may have heard the conventional wisdom saying that “*you are the average of the five people you spend the most time with*”. Implicitly, this is sometimes assumed to imply that your closest friends and peers have an effect on you, your personality, your success in life etc.

- a) **But can this causal claim actually be made?**

There are several problems with this statement, e.g.:

It is not clear, who affects who ($A \rightarrow B$ “your friends affect you” vs. $B \rightarrow A$ “you affect your friends”). If we are analyzing the occurrence of a trait and conclude that there is a correlation between us and our friends, we do not (necessarily) know who had the trait first and caused the trait to occur in others.

An important keyword in this context is spurious correlation – there are lots of studies showing strong correlations of variables with no causal relationship (e.g. the divorce rate in the state of Maine has a .99 correlation with the per capital consumption of margarine)

- b) **What other possibilities are there that would provide an explanation for any observed correlation?**

Correlations of traits can be results of other causes, such as:

- *Social influences (i.e. trait spill-overs)*
- *Co-occurrence of traits*
- *Selection of friends (grouping, based on traits)*

Exercise 5: Cronbach's Alpha & Explorative Analysis

In your survey, you have employed a multi-item construct with four items. Five participants took part in the survey.

- a) Based on their responses (below), what is the construct's Cronbach's alpha?

	Item 1	Item 2	Item 3	Item 4
Participant 1	1	5	4	3
Participant 2	2	4	3	4
Participant 3	2	2	2	2
Participant 4	3	4	4	3
Participant 5	2	5	5	4

The file `survey_data.csv` contains data from a survey with 8 constructs and a total of 31 items.

- b) What are the constructs' Cronbach's alpha values?
- c) Use a correlation plot to visualize the relations between all 31 items.
- d) For each construct, generate a new column with the mean of all of its items. Visualize the correlations between these aggregated construct values (e.g., correlation matrix with `ggally`).

➔ See files in ISIS for solution